

SAT Math Mastery · Week TP1I

SAT Trick Pack 1 — Part I

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics rewriting exponentials · parallel lines · inscribed right triangles · factor pairs · mixtures · rates and units · conjugate pairs · tangency · sequential transfers · chords · solids in cross-section · circular segments · data transformations

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Rewrite a shifted exponential with one base and read the initial value; a fractional base with a negative exponent grows.
- S2.** Recognise parallel, coincident and crossing lines from proportional coefficients.
- S3.** Spot a diameter among the sides of an inscribed triangle and use the right angle opposite it.
- S4.** Optimise a parameter over factor pairs, weighting each pair by its coefficients.
- S5.** Track one component through successive mixing stages and write one equation for the unknown.
- S6.** Combine a total-area equation and a total-volume equation using rates in the correct direction.
- S7.** Add fractions in conjugate pairs $(1 \pm a, 1 \pm a^2, \dots)$ and then invert.
- S8.** Use a zero discriminant (or centre-to-line distance) for tangency, and report the quantity actually asked for.
- S9.** Follow sequential fractional transfers between two groups with one variable.
- S10.** Convert units one factor at a time; time units are squared in an acceleration.
- S11.** Compute a chord from the centre-to-line distance and the radius.
- S12.** Reduce a solid to its axial cross-section; $\text{inradius} = \text{area}/\text{semiperimeter}$.
- S13.** Segment area = sector – central triangle; special right triangles give the central angle.
- S14.** Decide what a shift or an appended value does to the mean, median and standard deviation.

2 · REQUIRED FACTS

- $(\frac{1}{b})^{-u} = b^u$ and $(b^p)^q = b^{pq}$; the value at $x = 0$ of ab^x is a .
- Two lines with proportional x - and y -coefficients are parallel if the constants are not in the same proportion, and identical if they are.
- An angle inscribed in a semicircle is a right angle, so a chord equal to the diameter is the hypotenuse of a right triangle.
- In a mixture, only the *component* adds: $\text{amount} = \text{percent} \times \text{mass of that batch}$.
- A line is tangent to a circle exactly when the distance from the centre to the line equals the radius, equivalently when the substituted quadratic has discriminant 0.
- Rates may be given as “amount per area” or “area per amount”; these are reciprocals.
- Squaring a time-unit factor: $1 \text{ min}^2 = 3600 \text{ s}^2$.
- Adding a constant to every value shifts the mean and median by that constant and leaves the standard deviation unchanged; appending the mean leaves the mean and the sum of squared deviations unchanged.

3 · REQUIRED FORMULAS

$$\frac{1}{1-a} + \frac{1}{1+a} = \frac{2}{1-a^2}, \quad \frac{2}{1-a^2} + \frac{2}{1+a^2} = \frac{4}{1-a^4}, \quad \ell = 2\sqrt{r^2 - d^2}, \quad d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$

$$r_{\text{in}} = \frac{\text{Area}}{s}, \quad A_{\text{segment}} = \frac{\theta}{360^\circ} \pi r^2 - \frac{1}{2} r^2 \sin \theta, \quad V_{\text{sphere}} = \frac{4}{3} \pi r^3, \quad V_{\text{cone}} = \frac{1}{3} \pi R^2 h$$

4 · MINI THEORY SHEET

One base, then read. Write $(\frac{1}{8})^{(2-x)/3} = 2^{-(2-x)} = 2^{x-2}$ and split off the constant: $48 \cdot 2^{x-2} = 12 \cdot 2^x$. The coefficient is the value at $x = 0$.

Mixtures. For each stage write (percent)(mass) for the component only; pure additions contribute 100% of their mass. One equation: component after = percent after $\times$ mass after.

Tangency. Substitute the line into the circle; setting the discriminant to 0 gives an equation in the slope. Check whether the question asks for m or m^2 .

Cross-sections. A sphere inscribed in a cone is a circle inscribed in the axial triangle; its radius is area over semiperimeter.

Data. Work with the *sum of squared deviations*: a shift leaves it alone and appending the mean adds 0 to it, while the count grows.

Speed shortcuts.

Situation	Shortcut
$k \cdot b^{(c-x)/n}$ with $0 < b < 1$	rewrite with base $1/b$; this is growth, not decay; $f(0)$ is the coefficient
fractions with proportional left sides	same line or parallel: compare the constants after scaling
side of an inscribed triangle equals $2r$	right angle opposite it; Pythagorean triple for the missing side
$(pu + a)(qu + b)$ with fixed ab	$k = pb + qa$: pair the larger factor of ab with the larger coefficient
add pure component, then another alloy	one equation: all tin after = percent $\times$ total mass after
coverage rates in different forms	write litres per square metre for each; two equations: area and volume
$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \dots$	pair conjugates; the sum collapses step by step
line meets circle once	distance from centre to line = r ; solve for m^2 directly
fraction of A leaves, fraction of B returns	write A and B after each move in terms of x
acceleration in a new time unit	multiply by (time-unit factor) ²
chord from line and circle	d first, then $2\sqrt{r^2 - d^2}$; no intersection points needed
sphere in cone	slice through the axis; $r = \text{Area}/s$
chord and radius form a 30° - 30° - 120° pattern	read the central angle, then sector – triangle
shift then append the mean	mean shifts, SD falls, median must be re-checked

5 · GUIDED EXAMPLES

Example 1 — Pair the conjugates, then invert

For real t with $t^2 \neq 1$, $\frac{1}{1-t} + \frac{1}{1+t} = 5$. Find t^2 .

$$\frac{1}{1-t} + \frac{1}{1+t} = \frac{2}{1-t^2} = 5 \Rightarrow 1-t^2 = \frac{2}{5} \Rightarrow t^2 = \frac{3}{5}.$$

Example 2 — Chord from the distance, not the intersection points

Find the length of the chord cut from $x^2 + y^2 = 25$ by the line $x + y = 7$.

$$d = \frac{|0+0-7|}{\sqrt{2}}, \text{ so } d^2 = \frac{49}{2}. \text{ Chord} = 2\sqrt{25 - \frac{49}{2}} = 2\sqrt{\frac{1}{2}} = \sqrt{2}.$$

Example 3 — Inradius from area and semiperimeter

An isosceles triangle has sides 5, 5 and 6. Find the radius of its inscribed circle.

$$\text{Height} = \sqrt{25 - 9} = 4, \text{ area} = 12, \text{ semiperimeter} = 8. \quad r = \frac{12}{8} = \frac{3}{2}.$$

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined for $x \geq 0$ by

$$f(x) = 48 \left(\frac{1}{8} \right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x \geq 0$ and displays the value of $f(0)$ as a constant factor?

(A) $f(x) = 192 \cdot \left(\frac{1}{2} \right)^x$

(B) $f(x) = 12 \cdot 2^x$ ✓

(C) $f(x) = 48 \cdot 2^x$

(D) $f(x) = 12 \cdot \left(\frac{1}{2} \right)^x$

1. Write $\frac{1}{8} = 2^{-3}$. Then $\left(\frac{1}{8} \right)^{(2-x)/3} = 2^{-3 \cdot (2-x)/3} = 2^{-(2-x)} = 2^{x-2}$.

2. So $f(x) = 48 \cdot 2^{x-2} = 48 \cdot \frac{2^x}{4} = 12 \cdot 2^x$.

3. Check: $f(0) = 48 \left(\frac{1}{8} \right)^{2/3} = 48 \cdot \frac{1}{4} = 12$, which is the constant factor, and $f(1) = 48 \left(\frac{1}{8} \right)^{1/3} = 24 = 12 \cdot 2^1$.

4. The negative exponent $\frac{2-x}{3}$ reverses the effect of the fractional base, so f grows. Only $12 \cdot 2^x$ matches.

Answer: (B)

2.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{x}{6} + \frac{y}{4} = 2$$

Which of the following equations, when paired with the equation above, forms a system of two linear equations in x and y that has no solution?

(A) $4x - 6y = 40$

(B) $4x + 6y = 48$

(C) $4x + 6y = 40$ ✓

(D) $6x + 4y = 48$

1. Clear fractions in the given equation by multiplying by 12: $2x + 3y = 24$.
2. A system has no solution when the lines are parallel and distinct: the coefficients of x and y are proportional but the constants are not.
3. (A) $4x + 6y = 48$ is exactly $2(2x + 3y = 24)$: the same line, so infinitely many solutions.
4. (B) $4x + 6y = 40$: coefficients are $2\times$ those of $2x + 3y$, but $40 \neq 2 \cdot 24 = 48$. Parallel and distinct lines: no solution.
5. (C) $6x + 4y = 48$ and (D) $4x - 6y = 40$ have non-proportional coefficients (the determinants are $18 - 8 = 10$ and $-12 - 12 = -24$, both nonzero), so each gives exactly one solution.

Answer: (C)

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

Triangle ABC is inscribed in a circle with circumference 52π . Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?

(A) $\frac{10}{13}$ ✓

(B) $\frac{24}{13}$

(C) $\frac{5}{12}$

(D) $\frac{5}{13}$

1. Circumference $= 2\pi r = 52\pi$, so the radius is $r = 26$ and the diameter is 52.
2. $AC = 52$ equals the diameter, so $\overline{AC}$ is a diameter and the angle at B is a right angle (inscribed in a semicircle).
3. Pythagorean theorem: $BC = \sqrt{52^2 - 48^2} = \sqrt{(52 - 48)(52 + 48)} = \sqrt{400} = 20$.
4. The sides are 20, 48, 52; the shortest is $BC = 20$.
5. Ratio to the radius: $\frac{20}{26} = \frac{10}{13}$. (Dividing by the diameter would give $\frac{5}{13}$.)

Answer: (A)

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k ?

1. Expand: $(2x^2 + a)(3x^2 + b) = 6x^4 + (2b + 3a)x^2 + ab$. So $ab = 24$ and $k = 3a + 2b$.
2. For a fixed product, $3a + 2b$ is largest when the factor with the bigger weight takes the bigger share, and one factor is as small as possible. Integer pairs (a, b) with $ab = 24$ include $(1, 24)$, $(24, 1)$, $(4, 6)$, $(6, 4)$, $(8, 3)$ and the negative pairs.
3. $(a, b) = (1, 24)$: $k = 3 + 48 = 51$. $(a, b) = (24, 1)$: $k = 72 + 2 = 74$. Other positive pairs: $(2, 12)$ gives 30, $(12, 2)$ gives 40, $(3, 8)$ gives 25, $(8, 3)$ gives 30, $(4, 6)$ gives 24, $(6, 4)$ gives 26.
4. Negative pairs make k negative. The greatest value is $k = 74$.

Answer: 74

5.

SAT CHALLENGE · CALCULATOR

A metalworker melts x kilograms of an alloy that is 30% tin and mixes in 15 kilograms of pure tin. She then adds 25 kilograms of a second alloy that is 10% tin. The final alloy is 35% tin. What is the total mass, in kilograms, of the final alloy?

(A) 125 (B) 70

(C) 85 (D) 110 ✓

1. Tin after each stage: $0.30x$ from the first alloy, $+15$ from pure tin, $+0.10(25) = 2.5$ from the second alloy. Total tin: $0.30x + 17.5$.
2. Total mass of the final alloy: $x + 15 + 25 = x + 40$, and it is 35% tin: $0.30x + 17.5 = 0.35(x + 40)$.
3. $0.30x + 17.5 = 0.35x + 14 \Rightarrow 3.5 = 0.05x \Rightarrow x = 70$.
4. Final mass: $70 + 40 = 110$ kg. (Check: tin = $21 + 15 + 2.5 = 38.5$ and $0.35(110) = 38.5$.)
5. The value 70 is x , 85 is the mass after the pure tin is added.

Answer: (D)

6.

SAT CHALLENGE · CALCULATOR

A contractor finishes a 96-square-meter floor using two products. Primer P is applied in two coats, and each coat uses 125 milliliters per square meter. Sealant Q is applied in one coat and covers 5 square meters per liter. Every square meter receives exactly one of the two products, and the job used a total of 21.2 liters. How many square meters did the contractor cover with primer P?

1. Let p be the area of primer P and q the area of sealant Q, in square meters. Then $p + q = 96$.
2. Primer: two coats of 0.125 L per square meter is 0.25 L per square meter, so P uses $\frac{p}{4}$ liters. Sealant: 5 square meters per liter means $\frac{1}{5}$ liter per square meter, so Q uses $\frac{q}{5}$ liters.
3. Volume: $\frac{p}{4} + \frac{q}{5} = 21.2$. Multiply by 20: $5p + 4q = 424$.
4. Substitute $q = 96 - p$: $5p + 384 - 4p = 424 \Rightarrow p = 40$.
5. Check: $q = 56$, volume = $10 + 11.2 = 21.2$ L.

Answer: 40

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8 ?

(A) $\frac{5}{4}$

(B) $\frac{9}{5}$

(C) $\frac{1}{5}$ ✓

(D) $\frac{4}{5}$

1. Pair the first two terms: $\frac{1}{1-a} + \frac{1}{1+a} = \frac{(1+a) + (1-a)}{1-a^2} = \frac{2}{1-a^2}$.
2. Add the third term: $\frac{2}{1-a^2} + \frac{2}{1+a^2} = \frac{2(1+a^2) + 2(1-a^2)}{1-a^4} = \frac{4}{1-a^4}$.
3. Add the fourth term: $\frac{4}{1-a^4} + \frac{4}{1+a^4} = \frac{4(1+a^4) + 4(1-a^4)}{1-a^8} = \frac{8}{1-a^8}$.
4. So $\frac{8}{1-a^8} = 10 \Rightarrow 1-a^8 = \frac{4}{5} \Rightarrow a^8 = \frac{1}{5}$.
5. ($\frac{4}{5}$ is the value of $1-a^8$, not of a^8 .)

Answer: (C)

8.

SAT CHALLENGE · CALCULATOR

In the xy -plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2 ?

1. Substitute $y = mx + 10$: $x^2 + (mx + 10)^2 = 12 \Rightarrow (1 + m^2)x^2 + 20mx + 88 = 0$.
2. One intersection point means the discriminant is 0: $(20m)^2 - 4(1 + m^2)(88) = 0$.
3. $400m^2 - 352 - 352m^2 = 0 \Rightarrow 48m^2 = 352 \Rightarrow m^2 = \frac{352}{48} = \frac{22}{3}$.
4. Check by distance: $\frac{10}{\sqrt{1 + m^2}} = \frac{10}{\sqrt{25/3}} = \sqrt{12}$, the radius.
5. The question asks for m^2 , so the answer is $\frac{22}{3}$ (about 7.33), not m .

Answer: $\frac{22}{3}$

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

Study room A has x students and study room B has 72 students. One-third of the students in room A move to room B. Then one-fourth of the students who are now in room B move to room A. After both moves, the ratio of the number of students in room A to the number in room B is 5 to 7. How many students were in room A originally?

(A) 72

(B) 36 ✓

(C) 45

(D) 63

1. After the first move: A has $\frac{2}{3}x$ and B has $72 + \frac{x}{3}$.
2. Second move: one-fourth of B's students, $\frac{1}{4}\left(72 + \frac{x}{3}\right) = 18 + \frac{x}{12}$, go to A. A now has $\frac{2}{3}x + 18 + \frac{x}{12} = \frac{3}{4}x + 18$. B keeps $\frac{3}{4}\left(72 + \frac{x}{3}\right) = 54 + \frac{x}{4}$.
3. Ratio 5 : 7: $7\left(\frac{3}{4}x + 18\right) = 5\left(54 + \frac{x}{4}\right) \Rightarrow 5.25x + 126 = 270 + 1.25x \Rightarrow 4x = 144 \Rightarrow x = 36$.
4. Check: A: $36 \rightarrow 24 \rightarrow 24 + 21 = 45$; B: $72 \rightarrow 84 \rightarrow 63$; $45 : 63 = 5 : 7$.
5. The values 45 and 63 are the final counts, and 72 is the original count of room B.

Answer: (B)

10.

SAT CHALLENGE · CALCULATOR

A tram increases its speed at a constant rate from 54 kilometers per hour to 90 kilometers per hour in 20 seconds. What is the tram's acceleration, in meters per minute squared?

1. Convert the speeds to meters per minute: $54 \text{ km/h} = \frac{54,000}{60} = 900 \text{ m/min}$ and $90 \text{ km/h} = 1500 \text{ m/min}$.
2. Change in speed: $1500 - 900 = 600$ meters per minute.
3. Elapsed time in minutes: $20 \text{ s} = \frac{1}{3} \text{ min}$.
4. Acceleration $= \frac{600}{1/3} = 1800$ meters per minute squared.
5. Check in seconds: $\frac{25 - 15}{20} = 0.5 \text{ m/s}^2$, and $0.5 \cdot 60^2 = 1800$.

Answer: 1800

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

- (A) $300 \cdot 2^{m/18}$ (B) $1200 \cdot 2^{m/36}$
- (C) $900 \cdot 2^{m/18}$ (D) $1200 \cdot 2^{m/18}$ ✓

1. January 1, 2023 is 3 years after January 1, 2020, so $t = 3 + \frac{m}{12}$.
2. $A = 300 \cdot 4^{(3 + \frac{m}{12})/3} = 300 \cdot 4^{1 + \frac{m}{36}} = 300 \cdot 4 \cdot 4^{m/36} = 1200 \cdot 4^{m/36}$.
3. Change the base: $4^{m/36} = (2^2)^{m/36} = 2^{m/18}$. So $A = 1200 \cdot 2^{m/18}$.
4. Check: $m = 0$ gives $1200 = A(3) = 300 \cdot 4$. $m = 36$ gives $1200 \cdot 4 = 4800 = A(6) = 300 \cdot 16$.

Answer: (D)

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q . What is the length of $\overline{PQ}$?

(A) 6 ✓(B) $2\sqrt{21}$

(C) 8

(D) 3

1. Complete the squares: $(x + 4)^2 + (y - 1)^2 = 8 + 16 + 1 = 25$. Center $(-4, 1)$, radius $r = 5$.

2. Distance from the center to the line $5x + 12y - 44 = 0$: $d = \frac{|5(-4) + 12(1) - 44|}{\sqrt{25 + 144}} = \frac{52}{13} = 4$.

3. The perpendicular from the center bisects the chord, forming a right triangle with legs $d = 4$ and $\frac{PQ}{2}$ and hypotenuse $r = 5$: $\frac{PQ}{2} = \sqrt{25 - 16} = 3$.

4. $PQ = 2 \cdot 3 = 6$. (3 is the half-chord; $2\sqrt{21} = 2\sqrt{25 - 4}$ comes from using $d = 2$; $8 = 2d$.)

Answer: (A)**13.**

SAT CHALLENGE · NO CALCULATOR NEEDED

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)

1. Slice through the axis: the cross-section is an isosceles triangle with base 10, height 12 and slant sides $\sqrt{5^2 + 12^2} = 13$. The sphere appears as the circle inscribed in this triangle.

2. Area = $\frac{1}{2}(10)(12) = 60$. Semiperimeter $s = \frac{13 + 13 + 10}{2} = 18$. Inradius $r = \frac{60}{18} = \frac{10}{3}$.

3. Sphere volume: $\frac{4}{3}\pi \left(\frac{10}{3}\right)^3 = \frac{4000\pi}{81}$. Cone volume: $\frac{1}{3}\pi(5^2)(12) = 100\pi$.

4. Fraction = $\frac{4000/81}{100} = \frac{40}{81}$.

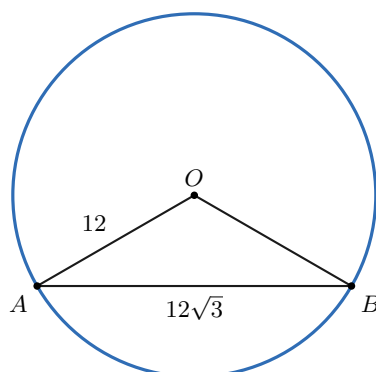
5. Check: the center is at height $\frac{10}{3}$ on the axis and its distance to a slant side is $\frac{5(12 - \frac{10}{3})}{13} = \frac{10}{3}$.

Answer: $\frac{40}{81}$

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB ?



(A) $24\pi - 36\sqrt{3}$

(B) $48\pi - 36\sqrt{3}$ ✓

(C) 48π

(D) $96\pi + 36\sqrt{3}$

- Triangle OAB has $OA = OB = 12$. Drop the perpendicular from O to $\overline{AB}$: it bisects the chord into two segments of length $6\sqrt{3}$.
- In each half-triangle, $\sin\left(\frac{\angle AOB}{2}\right) = \frac{6\sqrt{3}}{12} = \frac{\sqrt{3}}{2}$, so $\frac{\angle AOB}{2} = 60^\circ$ and $\angle AOB = 120^\circ$.
- Sector area: $\frac{120}{360}\pi(12)^2 = 48\pi$.
- Triangle area: $\frac{1}{2}(12)(12)\sin 120^\circ = 72 \cdot \frac{\sqrt{3}}{2} = 36\sqrt{3}$.
- Segment = sector – triangle = $48\pi - 36\sqrt{3}$. (Option 48π omits the triangle; $96\pi + 36\sqrt{3}$ is the major segment.)

Answer: (B)

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has six values: 2, 3, 5, 12, 18, 20. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.

- I. The mean is 4 greater.
- II. The standard deviation is smaller.
- III. The median is 4 greater.

Which of the statements must be true?

(A) I, II and III

(B) I only

(C) I and II only ✓

(D) II and III only

1. Original: sum = 60, mean = 10, median = $\frac{5+12}{2} = 8.5$. After adding 4 to each value: mean 14, median 12.5, values 6, 7, 9, 16, 22, 24.
2. I: the appended value is the new mean, 14, so the mean stays 14, which is 4 more than 10. True.
3. II: adding 4 to each value does not change the sum of squared deviations (306). Appending the mean adds 0 to it, while the number of values grows from 6 to 7, so the standard deviation falls. True.
4. III: the seven values in order are 6, 7, 9, 14, 16, 22, 24, so the median is 14, not $8.5 + 4 = 12.5$. False.
5. Statements I and II only.

Answer: (C)